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UNIT-5.1:

BODY MECHANICS, MOVING AND LIFTING

Body mechanics

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- **Body Mechanics:** the purposeful and coordinated use of body parts and positions during activity.
- Use of proper body mechanics is important during *lifting and moving a patient as this prevent injury to the muscles, joints, and tendons.*
- Proper body mechanics are as important to the nurse as to the client

Principles of body mechanics

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1. Less energy is used if all body parts are balanced appropriately
2. The greater the base of support, the more stable the body. Force can be applied or resisted more effectively when the base of support can be enlarged in the direction of the force
3. Pelvic tilt(contraction of the abdominal and gluteal muscles to stabilize the pelvis) before activity helps to protect the lower back from strain and injury

Principles cont...

4. Less energy is needed to keep an object moving than to initiate movement
5. Moving an object on a level surface requires less effort
6. Reducing friction between the object moved and the surface on which it is moved requires less energy
7. Holding an object close to the body requires less energy than holding it farther away

Principles cont...

8. Muscle strain can be avoided by using the strong leg muscles when lifting, pushing, and pulling
9. Smooth, continuous movements are easier and safer than sudden, sharp, or uncontrolled movement
10. Using rhythmic movements at a normal speed requires less energy

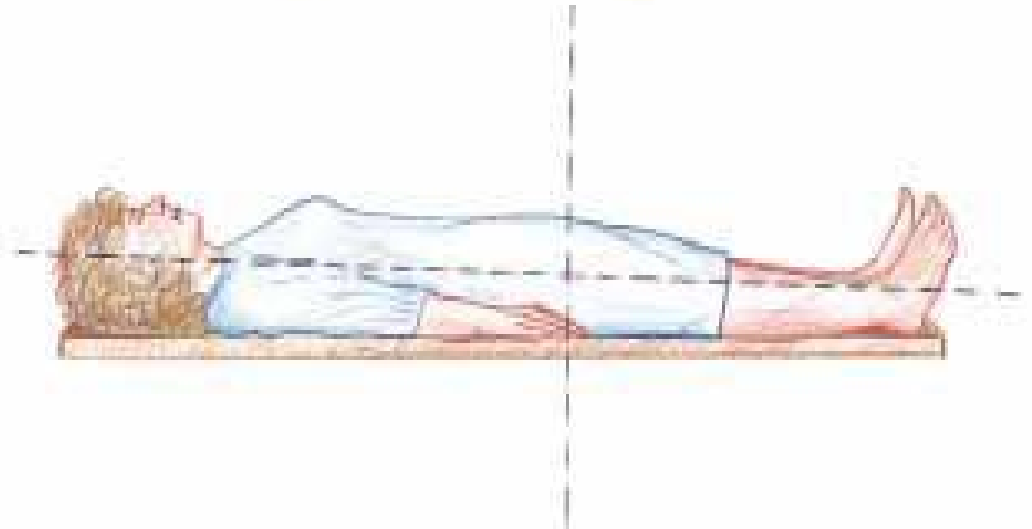
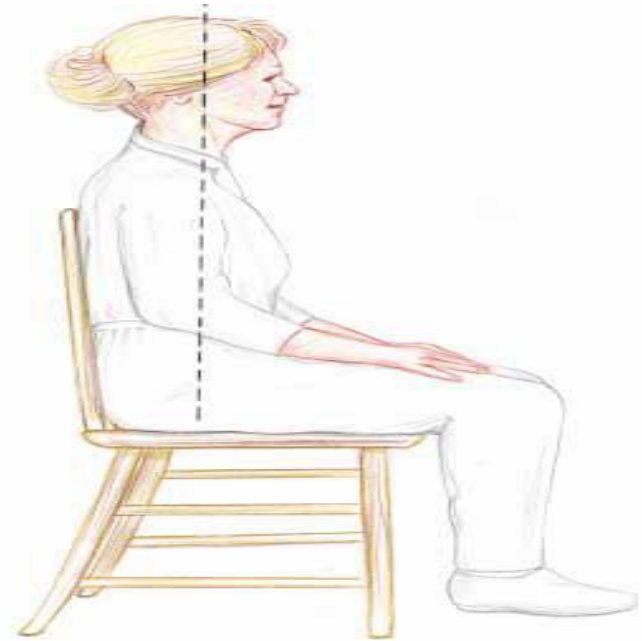
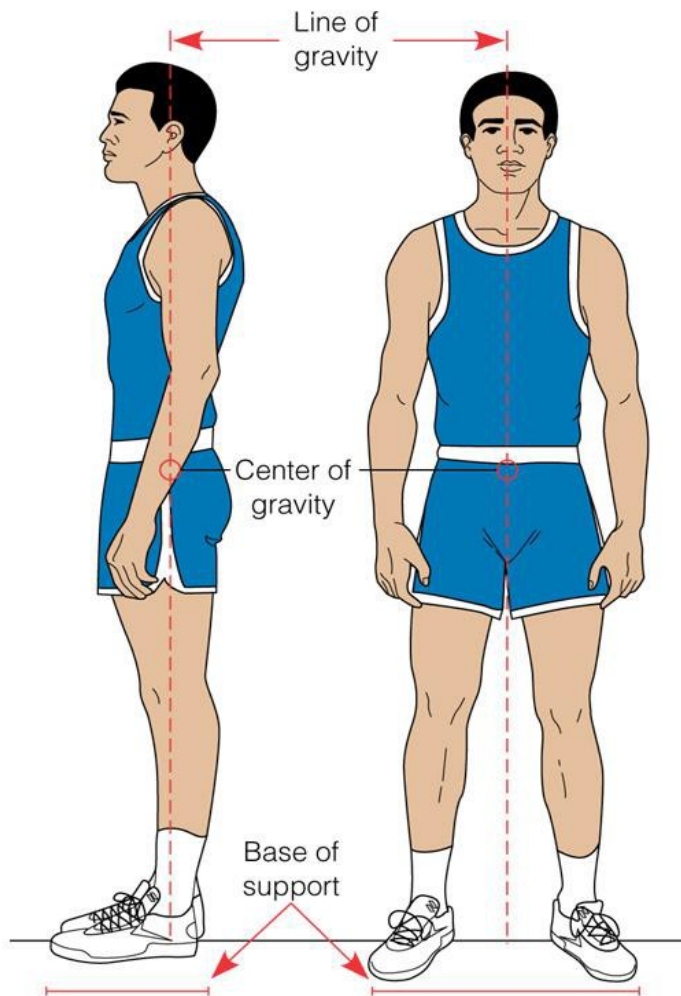
Cont'd...

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- Principles underlying proper body mechanics involve three major factors : ***center of gravity, base of support, and line of gravity.***
- **Center of Gravity** : the point where the center of mass of the body is located.
 - The person's center of gravity located in the pelvic area
- **Base of Support** : person's feet provide the base of support.
 - The wider the base of support, the more stable the object within limits.

- **Line of gravity(or the gravity plane)** : an imaginary vertical (up and down) line that passes through the top of the head, the center of gravity, and the base of support.
 - This is the direction of gravitational pull (from the top of the head to the feet).
 - For highest efficiency, this line should be straight from the top of the head to the base of support, with equal weight on each side.

Cont'd...



- ***Proper standing body alignment is X-zed by the following:***
 - Head upright
 - Face forward
 - Shoulders squared
 - Back straight
 - Abdominal muscles tucked in
 - Arms straight at side
 - Hands palm forward
 - Legs straight
 - Feet forward

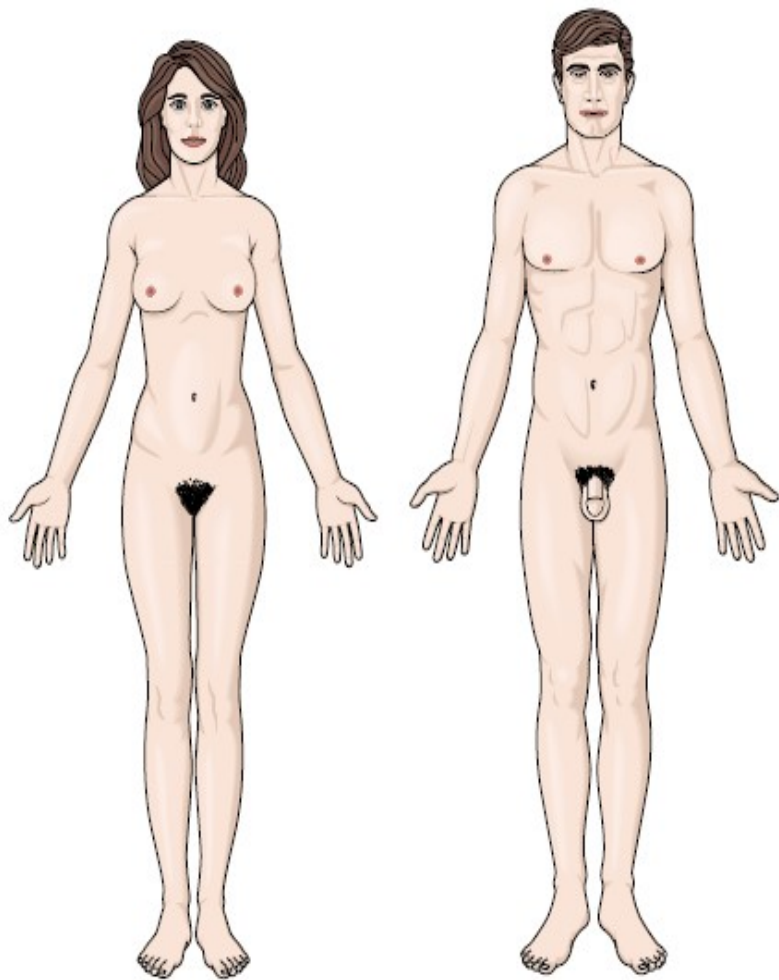


Figure 34-2 Proper Alignment and Posture: Standing Male and Female

Cont'd...



Proper Sitting Posture and Line of Gravity

- The benefits of proper alignment and posture
 - Client comfort
 - Prevention of contractures;
 - Promotion of circulation
 - Less stress on muscle, tendons, nerves, and joints
 - Prevention of foot drop (plantar flexion)

EXERCISE

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- A type of physical activity defined as a planned, structure and repetitive bodily movement done to improve or maintain one or more components of physical fitness



TYPES OF EXERCISE

□ ISOTONIC

- Dynamic exercise in which the muscle shortens to produce contraction and movement
- E.g. Running, walking, swimming, cycling



TYPES OF EXERCISE CONT'D..

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□ ISOMETRIC

- Are those in which there is a change in muscle tension but **NO CHANGE** in muscle length
- **E.g. Tensing, extending and pressing exercises**



TYPES OF EXERCISE CONT'D..

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❑ ISOKINETIC

- Involves muscle contraction or tension against a resistance



TYPES OF EXERCISE CONT'D..

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❑ AEROBIC EXERCISE

- Activity during which the amount of oxygen taken into the body is greater than that used to perform the activity



BENEFITS OF EXERCISE

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- Increases joint flexibility, tone and ROM
- Bone density is maintained
- Increases cardiac output and perfusion
- Prevents pooling of secretions in the lungs
- Improves appetite and facilitate peristalsis
- Elevates the metabolic rate
- Prevents stasis of urine
- Produces a sense of well-being

IMPAIRED PHYSICAL MOBILITY

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Complications of IMMOBILITY

- Contractures, atrophy and stiffness
- Foot drop
- DVT
- Hypostatic pneumonia
- Pressure ulcers, skin breakdown, reduced skin turgor

Compns of immobility Cont'd..

- Muscle atrophy
- Osteoporosis
- Dependent edema
- Urine stasis
- Constipation



Ambulation (assisting a pt to walk)

- Client ambulation (assisted or unassisted walking) is encouraged soon after the onset of *illness or surgery* to *prevent the complications of immobility*
- While the client is walking,
 - observe for signs of exertion (*including diaphoresis, shortness of breath ,or weakness*).
 - *It is also important to assess for the presence of orthostatic hypotension in order to prevent falls*

Ambulation cont...

- Immobile clients should be prepared for ambulation through *pre-ambulatory exercises* in order to strengthen and tone muscles(i.e. clients should be directed to gently contract and release the leg muscles, especially quadriceps femoris, several times a day)

Ambulation cont...

□ Purposes of ambulation

- ❖ To prevent complications following debilitating illness or operation (respiratory stasis, urinary retention, constipation, thrombosis or thrombo-embolism, etc..)
- ❖ To strengthen weak muscles/prevents muscle atrophy
- ❖ To speed up and restore independence for self care
- ❖ To increase the patient's confidence for self care

Ambulation cont...

1. **Assisting patient with assistive devices**

□ Assistive device is a material used to support client/patient who unable to walk independently

□ **Common types**

- Gait belt
- Cane
- Walker
- Crutch

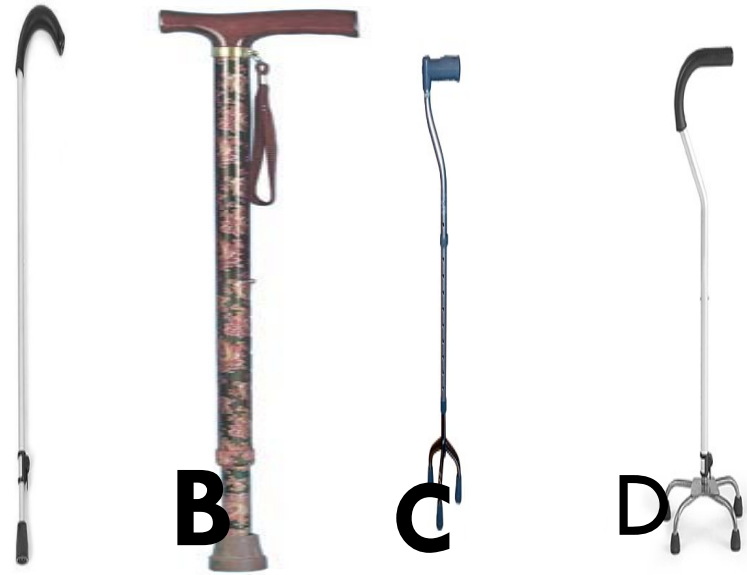
a. Gait belt

- Is an assistive used during simple assisted ambulation
- Aid ambulation of patient with one body part weakness



b. Cane

- Used by clients who can bear weight on both legs but have some weakness in one leg or hip
- It supports patient during walking



- A. Standard Cane
- B. T-handle Cane
- C. Tripod Cane
- D. Quad (Quadruped) Cane

c. Walker

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- A walker is a waist-high metal tubular device with a handgrip and four legs X-zed by the presence of **rubber tips on all four legs** or have **wheels on the two front legs**
- **Purpose**
 - To provide extra support, sense of security, and independence
- **Indication**
 - Weakness in lower Limbs
 - Disability' Postoperative (as needed)

□ Indication

- Weakness in lower Limbs
- Disability
- Postoperative (as needed)



d. Crutch

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- **Crutch** is walking aids made of wood or metal in the form of a shaft which reaches from the ground to the client's axilla.
- **Crutch walking;** is one of the patient ambulation techniques which helps the client to stand and walk with the help of Crutch
- Used to:
 - ▣ Reduces anxiety and promotes client's autonomy
 - ▣ Improve muscle strength

Types of Crutch



Axillary crutch



Forearm crutch



Platform crutch

1. Axillary crutches

ADVANTAGES:

- ❑ Improve balance & lateral stability.
- ❑ Provide for functional ambulation.
- ❑ Easily adjusted.
- ❑ Inexpensive when made of wood.
- ❑ Can be used for stair climbing.

DISADVANTAGES:

- ❑ Crutches are awkward in small areas.
- ❑ Safety of the user may be compromised when ambulating in crowded areas.
- ❑ The tendency of the patients to lean on the axillary bar.
- ❑ This pressure creates the potential for damage to nervous & vascular structures in the axilla.

Parts of the Axillary crutches

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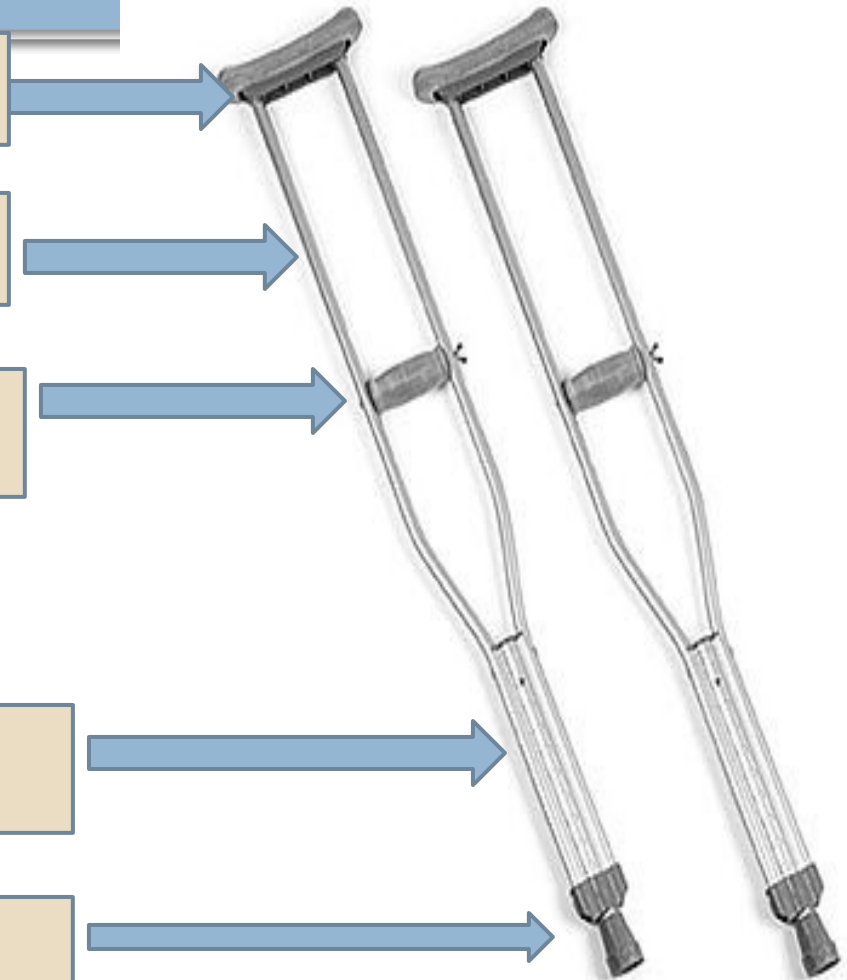
Axillary Bar & Pad

Double Uprights

Hand Piece

Single Leg

Rubber Suction Tip



2. Forearm Crutches

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ADVANTAGES:

1. Allows use of hands without the crutches becoming disengaged.
2. Easily adjusted & allow functional stair-climbing activities.
3. Many patients feel they are more cosmetic & they fit more easily into an automobile.
4. They are also the most functional type of crutch for stair-climbing activities.

DISADVANTAGES:

1. Provide less lateral support owing to the absence of an axillary bar
2. The cuffs may be difficult to remove.
3. These crutches are more costly than wooden axillary crutches.

Parts of a Forearm Crutches

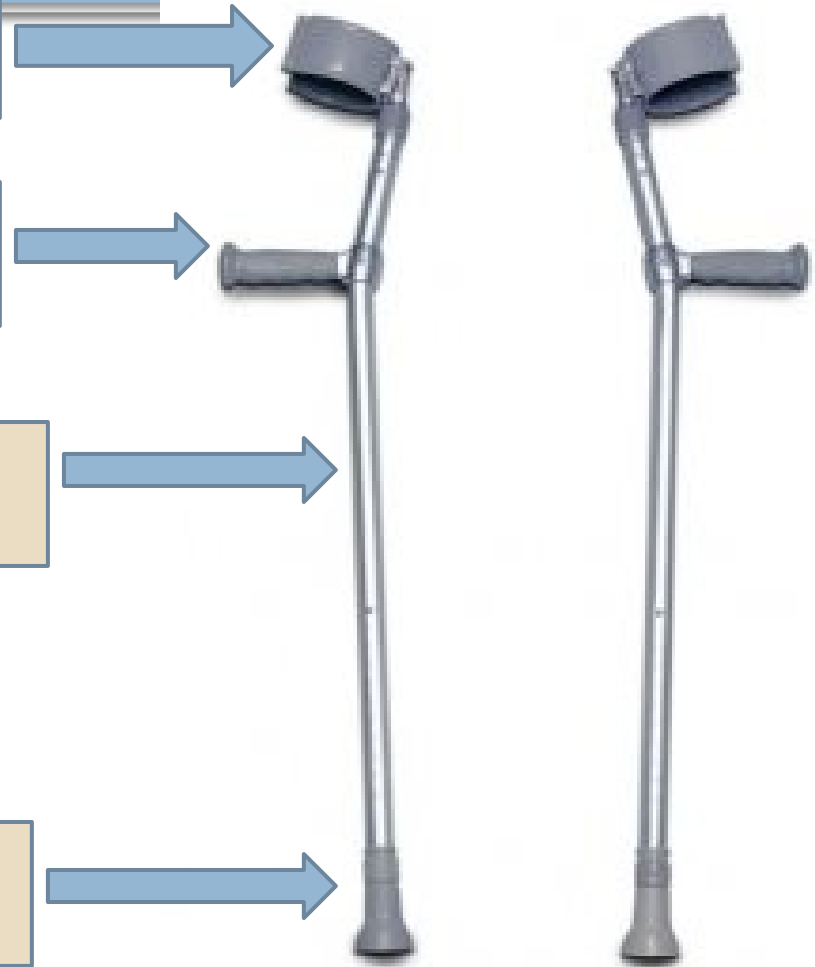
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Forearm Cuff

Hand Grip

Single Upright

Rubber Suction Tip



3. Platform Crutches

- Are also referred as **Forearm Rests** or **Troughs**
- A special crutches that is used when weight bearing is contraindicated through the wrist and hand.
- For e.g.: Patients with Arthritis



Gait Used with crutches

- **Assisting a Client with Crutch Walking**
- ***Equipment***
 - One pair of crutches
 - Measuring tape
 - Gait belt (optional)

Gait Used with crutches cont...

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Procedure

1. Inform client that you will be assisting with ambulation using crutches.
2. **Assess client for strength, mobility, range of motion, visual acuity, perceptual difficulties, and balance.**
3. **Adjust crutches to fit the client. With the client supine, measure from the heel to the axilla.**

With the client standing, set the crutch position at a point 4 to 5 in. lateral to the client and 4 to 6 in. in front of the client. The crutch pad should fit 1.5 to 2 in. below the axilla. The hand grip should be adjusted to allow for the client to have elbows bent at 30° flexion.



Crutches cont'd...

4. Lower the height of the bed.
5. **Dangle the client at the side of bed for several minutes. Assess for vertigo.**
6. **Instruct client on method to hold the crutches; that is, with elbows bent 30° and pad 1.5 to 2 in. below the axilla. Instruct client to position crutches lateral to and forward of feet. Demonstrate correct positioning.**
7. **Apply the gait belt around the client's waist if balance and stability are impaired.**
8. Assist the client to a standing position with crutches. Support as needed

Gait Used with crutches cont...

i. Four-Point Gait

9. Position the crutches 4.5 to 6 in. to the side and in front of each foot. Move the right crutch forward 4 to 6 in. and move the left foot forward, even with the left crutch (Figure A).

- Move the left crutch forward 4 to 6 in. and move the right foot forward, even with the right crutch (Figure B). Repeat the four-point gait.

A.



B



Gait Used with crutches cont...

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ii. Three-Point Gait

10. Advance both crutches and the weaker leg forward together 4 to 6 in. Move the stronger leg forward, even with the crutches. Repeat the three-point gait.



Gait Used with crutches cont...

iii. Two-Point Gait

11. Move the left crutch and right leg forward 4 to 6 in. Move the right crutch and left leg forward 4 to 6 in. Repeat the two-point gait.

Gait Used with crutches cont...

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iv. Swing-Through Gait

12. Move both crutches forward together 4 to 6 in. Move both legs forward together in a swinging motion, even with the crutches. Repeat the swing-through gait.
13. Set realistic goals and opportunities for progressive ambulation using crutches.
14. **Consult with a physical therapist for clients** learning to walk with crutches.



Lifting & Moving a Patient

- **Lifting** : moving an object or person against gravity.
- **Moving** : displacing an object, person, or parts of the person's body from its original position
- **Purposes of Moving & lifting Pt**
 - To assist a patient who is un able to move himself
 - To prevent fatigue & injury
 - To maintain good body pasture (alignment)
 - To stimulate circulation
 - To prevent venous thrombosis

Lifting the patient

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a. **Dangling:** is sitting on the side of the bed with the feet hanging down

□ **Purpose**

- ▣ To prepare patient before walking ,moving to chair or wheelchair or performing others
- ▣ To relive pressure in case of pulmonary edema

□ **Indication**

- Moving patient out of bed

□ **Contraindication**

- Unconscious patient; Spinal injury

Dangling cont...

□ **Precaution**

- ❖ Do not leave the patient alone when dangling.
- ❖ If the patient becomes dizzy lie him down.
- ❖ Have the patient cough, deep breathe, and exercise their leg muscles when dangling
- ❖ Check the person's pulse and respirations

□ **Equipment**

- Turn sheet or draw sheet
- Screen

Dangling cont....



Assist the client from a supine to a seated position



Help the client move into the dangling position

Procedure

Dangling cont....

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1. Greet the patient and explain the procedure
2. Perform hand hygiene
3. Collect the necessary material
4. Provide privacy
5. Assess the patient condition
6. Position yourself and client appropriately before performing the move
 - a. Assist the client to a lateral position facing you
 - b. Raise the head of the bed slowly to its highest position
 - c. Position the clients feet and lower legs at the edge of the bed
 - d. Stand beside the client's hips and face the far corner of the bottom of the bed

Procedure cont'd...

47 7. Move client to sitting position

- a. Place one arm around the client shoulder and the other arm beneath both of the client thighs near the knee
- b. Tighten your gluteal , abdominal, leg and arm muscle
- c. Lift the client thighs slowly
- d. Pivot on your feet in the desired direction facing the foot of the bed while pulling the client feet and legs off the bed
- e. Keep supporting the client until client is well balanced and comforted
- f. Assess vital signs as indicated

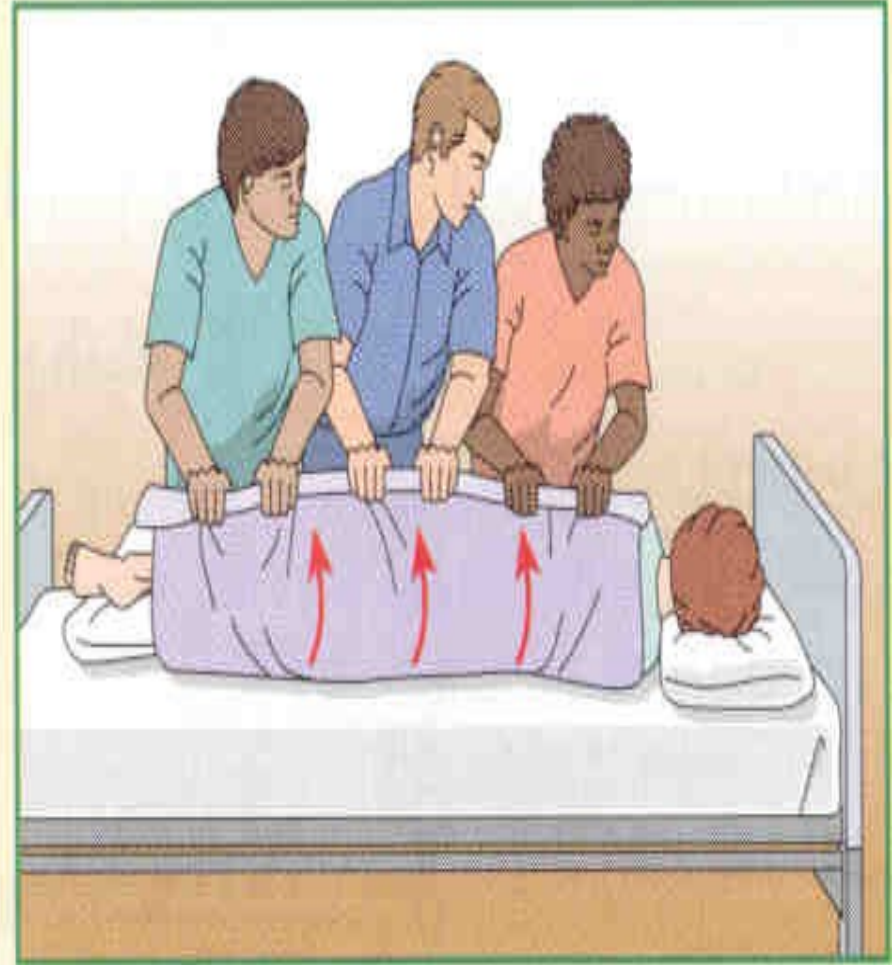
8. Document all relevant information

b. Log rolling

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- It is a technique used to turn a patient whose body must at all times be kept in a straight alignment.
- Aimed to turn a patient to the side of bed
- **Indication**
 - Spinal injury
- **Note:** logrolling is accomplished by two or three nurses working in a coordinated fashion

Log rollina cont...



Log rolling with three persons

Log rolling cont...

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Procedure

1. Wash your hands
2. Inform client of reason for the move and how to assist (if able).
3. Elevate hospital bed to high position.
4. Using one or more staff members, place a turn/draw sheet under the client's back and head.
5. The lead nurse tells the client and other personnel the direction of the move.

6. One person stands on each side of bed. The

- ▣ lead nurse gives the signal for the move. The
- ▣ staff member on side of the bed in the direction
- ▣ of the move holds the turn/draw sheet to guide
- ▣ the move. The second staff member applies
- ▣ gentle pressure on client's back toward the
- ▣ direction of the move, assisting client to roll

Procedure cont'd...

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7. Tuck pillows at client's back and abdomen.
8. Assess the client for comfort and proper alignment.
9. Elevate side rails and lower the bed height.
10. This procedure can be reversed to reposition clients on their backs.



Moving the patient up in bed

- Moving the patient to the head side of the bed.

Purpose

- To prevent the patient sliding down the bed there by maintaining good body alignment
- To put the patient in sitting or semi-sitting position
- It is done when Patient slides to the foot of the bed

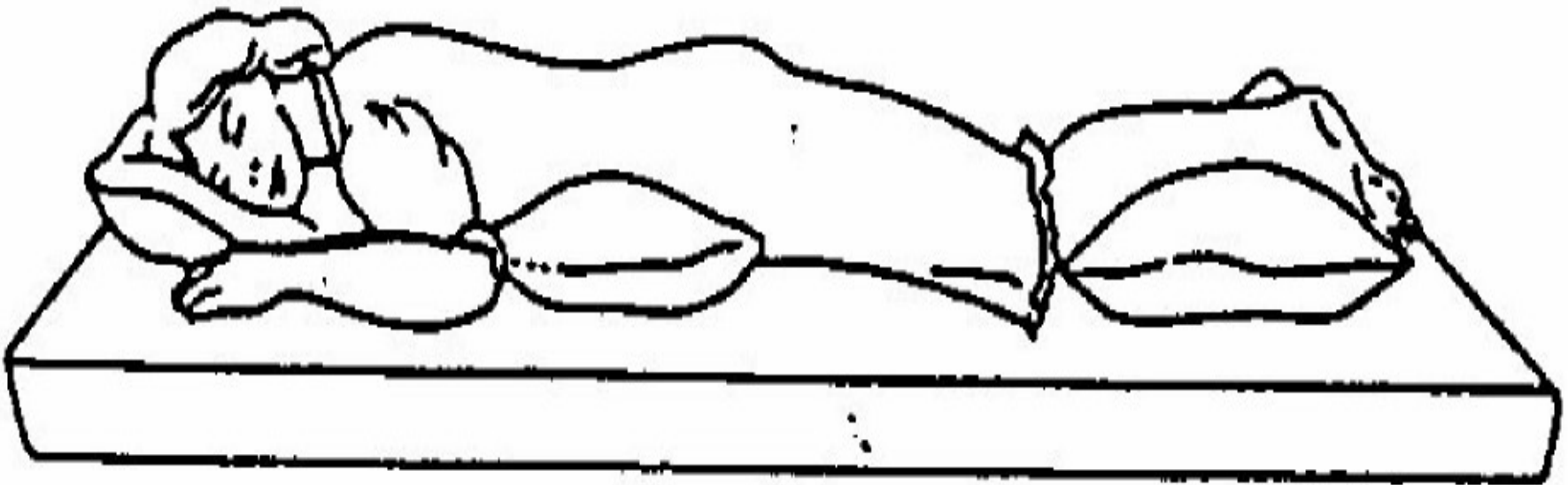
Positioning the patient

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- **Positioning is turning or putting the patients in a proper body alignment for the purpose of:**
 - ❖ Relieving pressure on various parts
 - ❖ Preventing formation of deformity
 - ❖ Improving circulation
 - ❖ Preserving muscle function as different muscle group's contract and relax
 - ❖ Providing comfort, support, and good body alignment
 - ❖ Making the patient ready for different procedures

1. Front lying (Prone) Position

- Prone position is putting the patient in a flat on the abdomen, legs extended, feet over the edge of the mattress, and toes pointing to the floor



Indications

- Used in post operative case of tonsillectomy
- in case of burn over back
- To prevent bed sore
- To provide back massaging and lung, hip assessment

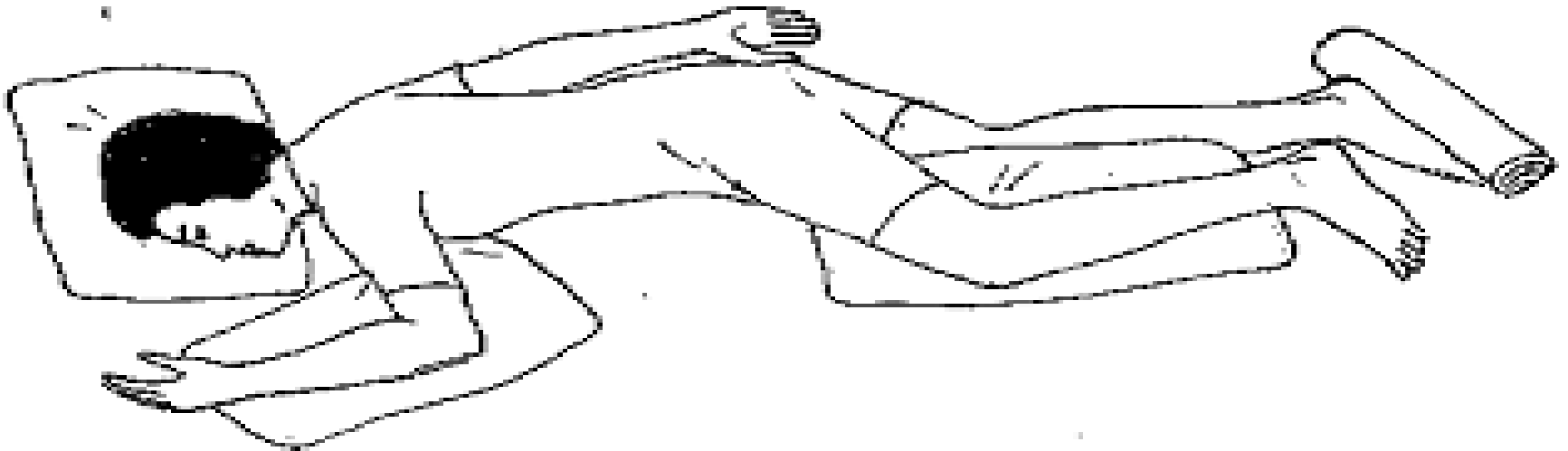
□ **Contraindication**

- Cervical –spine fracture
- Respiratory impairment/breathing difficulties
- Foot drop
- Pregnant women
- Clients with abdominal incisions

2. Semi-prone/Sims' position

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- Semi-prone position putting or assisting patients with upper arm flexed at shoulder and elbow; lower arm positioned behind client and both legs flexed in front of client with more flexion in upper leg either of body side



□ Purpose

- To promotes drainage from mouth
- To prevents aspiration
- Comfortable for sleeping.
- Promotes comfort especially in pregnant clients

Sims' position cont'd...

□ Indication

- For rectal examination
- Pressure sore on the buttocks/sacrum and hips

□ Contraindication

- Lumbar lordosis
- Foot drop
- Client with leg injuries or arthritis

- **Supine position** is putting patient in back lying often with a small pillow to support the head and shoulder
3. Supine position



Purpose

- Promote comfort
- To help healing after certain abdominal operations

Supine position cont'd...

□ Indication

- ▣ After abdominal, chest and neck surgery
- ▣ For physical examination of anterior part of the body
- ▣ Usual position for the patient

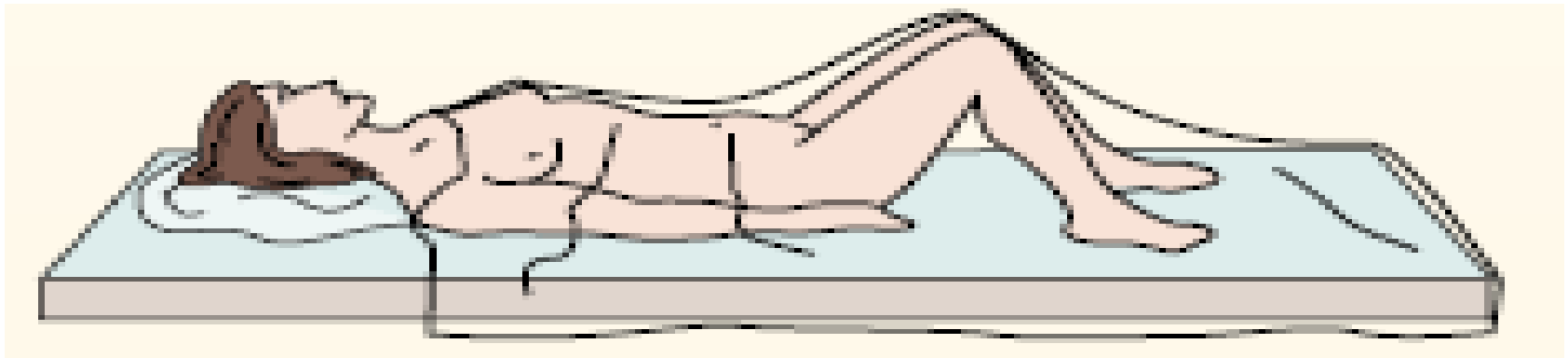
□ Contraindication

- ▣ Spinal injury
- ▣ Cardiac patient (CHF)
- ▣ Breathing impairments
- ▣ Pressure sore (buttock, scum, heel and shoulder)

3. Dorsal recumbent

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- Is putting patient in back lying position with knees are flexed and the soles of the feet flat on the bed
- **Purpose**
 - Promote comfort, For visualize the perineum
 - To insert urinary catheter
 - To relief pressure from ileum, knee and ankle



Dorsal recumbent cont'd...

□ Indication

- ❖ Rectal , vulval and vaginal examination
- ❖ Pelvic surgical procedures
- ❖ Vaginal douche
- ❖ Perineal care
- ❖ Catheterization
- ❖ Supra-pubic puncture

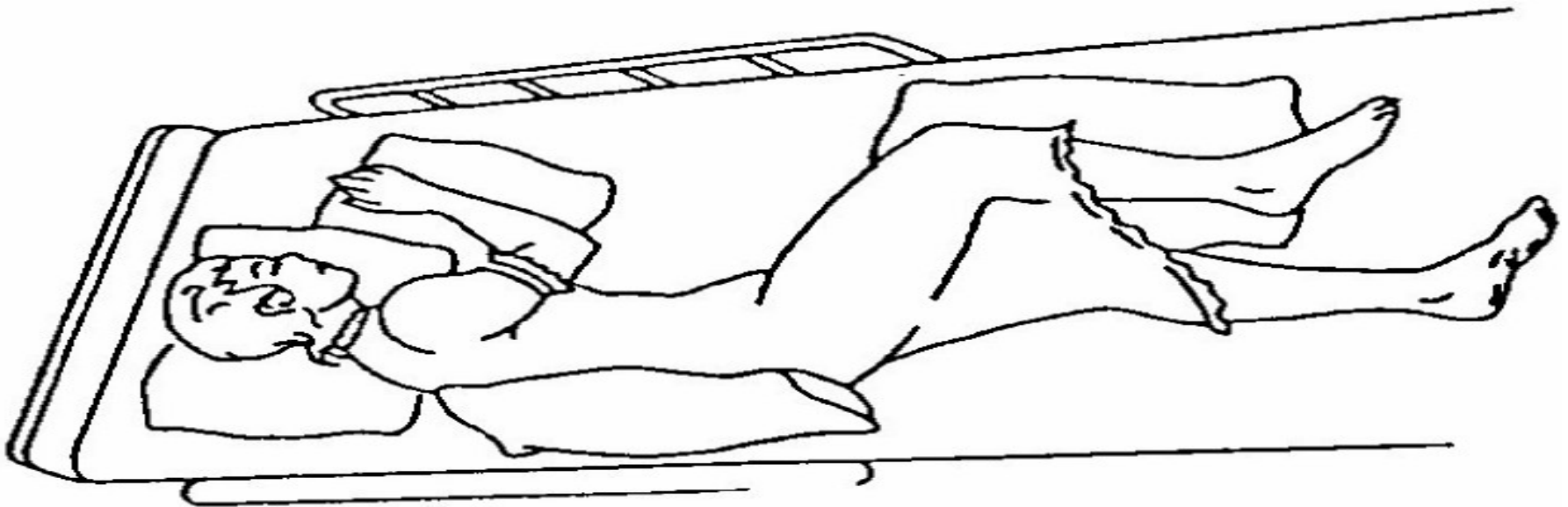
Dorsal recumbent cont'd...

□ **Contraindication**

- ❖ Spinal injury
- ❖ Cardiac patient (CHF)
- ❖ Breathing impairments
- ❖ Pressure sore (buttock, scrum, heel and shoulder)

4. Lateral recumbent (on either side)

- Lateral recumbent position is putting patient on either of the side with the legs flexed at knee (The upper leg is more flexed than the lower leg)



Lateral recumbent cont'd...

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□ Purpose

- To perform back care
- To relieve pressure on sacrum and heels
- To perform enema
- To take rectal body temperature
- To insert suppositories

Lateral recumbent cont'd...

□ Indication

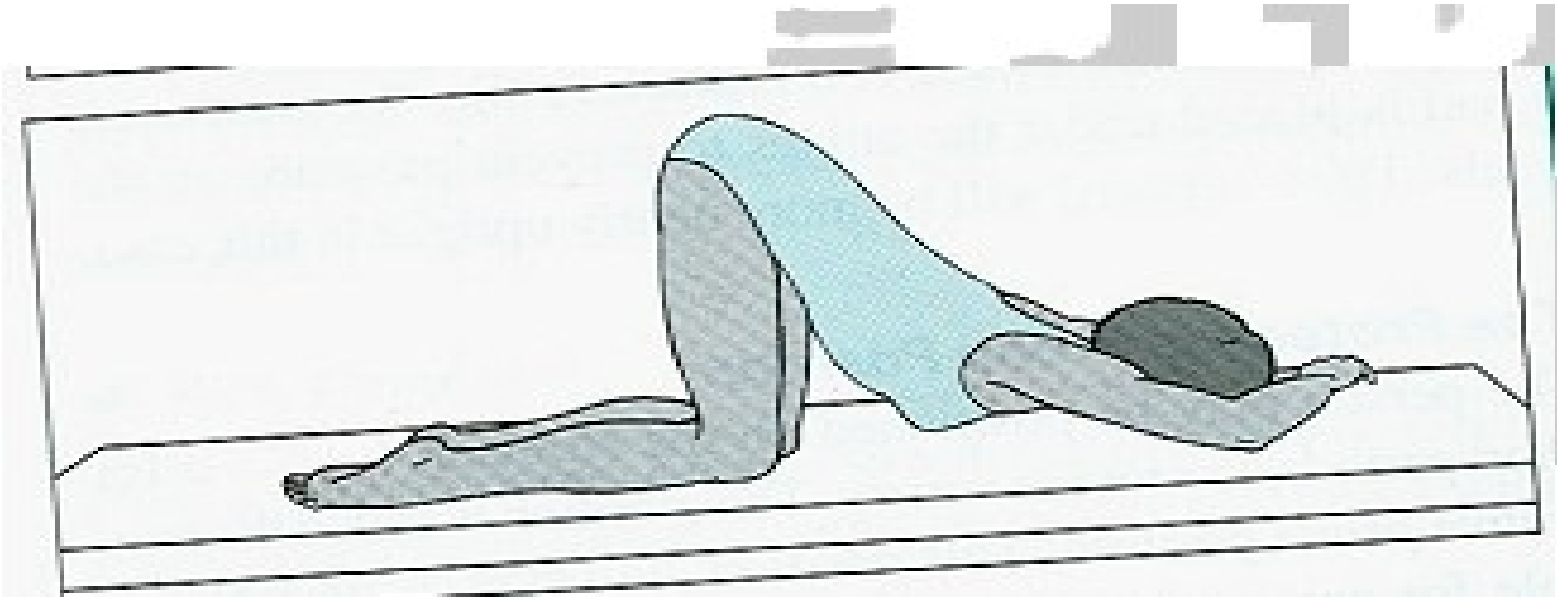
- Enema and colonic irrigation
- Pressure sore on heel and sacrum
- Rectal examination
- To measure rectal temperature

□ Contraindication

- Arm and rib fracture
- Spinal injury
- Flank injury

5. Knee chest

- Is putting a patient on the Knee and the chest with the head turned one side, arms above the head and one cheek on a pillow



□ Indication

- Sigmoidoscopic examination
- Vaginal and rectal examination
- Cord Prolapse
- Retroverted uterus

□ Contraindication

- Cardio-pulmonary problem
- Upper arm, spine and ribs fracture
- Increased intra-cranial pressure

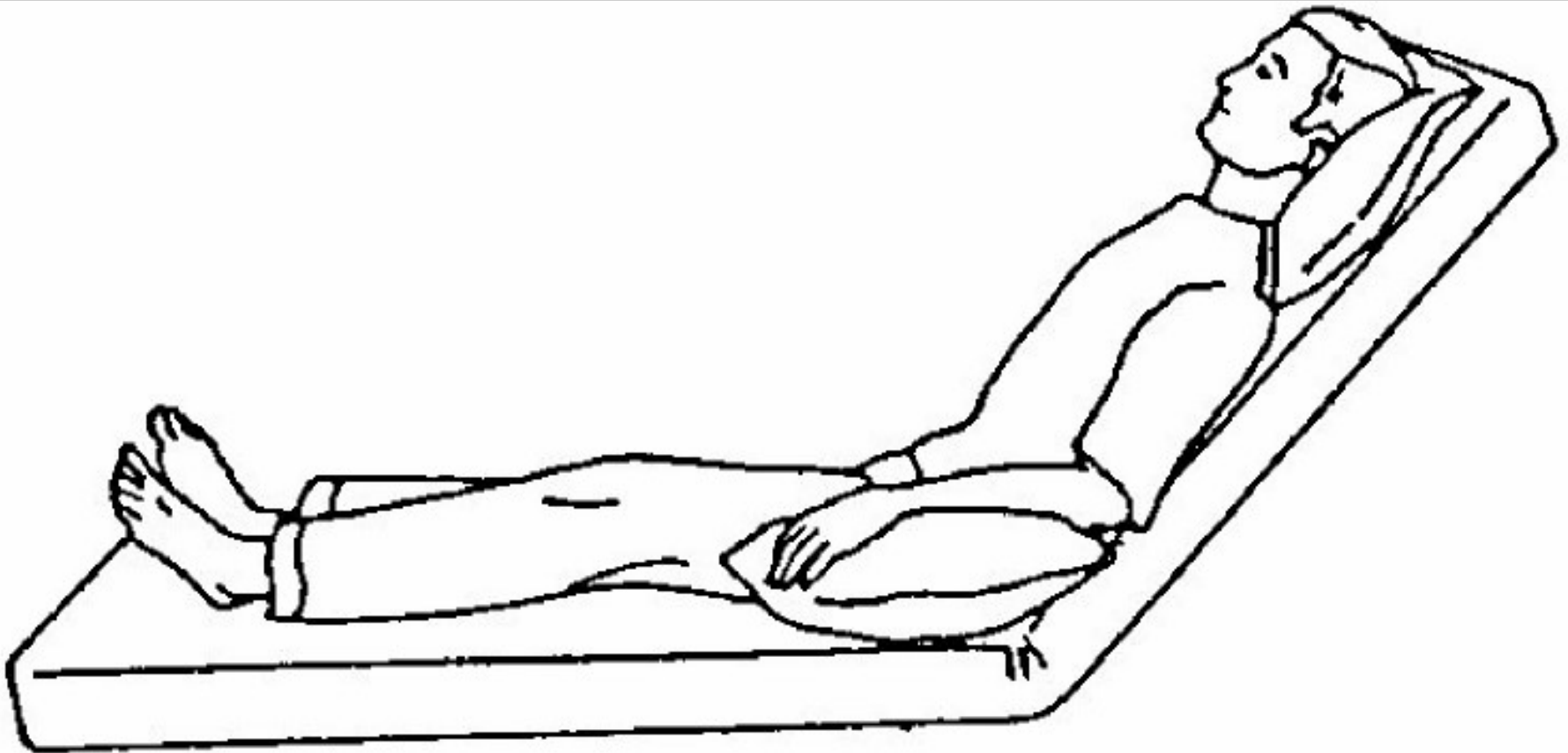
6. Fowler's position

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- Fowler's position is sitting position in which the head is elevated at different angle (15-90)° angle and may have knees either bent or straight
- **Type**
 1. **High Fowler's** position is when the patient's head is raised 80-90 degrees,
 2. **Semi-Fowler's** position is when the patient's head is elevated 30-45 degrees.
 3. **Low Fowler's** position is when the head of bed is elevated 15-30 degrees
 4. **Fowler's** which is 45-60 degrees

Fowler's position cont'd...

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Proper fowler's position

AGHMC; By Amana A. 09/23/2026

Procedure

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- a. To relieve dyspnea
- b. To improve circulation
- c. To prevent thrombosis
- d. To prevent aspiration during the introduction of feeding tubes
- e. To facilitate drainage from abdomen and pelvic cavity post operatively
- f. To relax the muscle of the abdomen, back and thighs
- g. To relive tension on abdominal suture
- h. To promote comfort
- i. Increase comfort during eating
- j. To relieve edema of the chest and abdomen

□ Indication

- ▣ Cardio-pulmonary problem (Respiratory distress, CHF, pulmonary edema..)
- ▣ Increased intra abdominal pressure
- ▣ Thrombosis
- ▣ Abdominal, back and thigh muscle strain
- ▣ Nasal or oral passageway procedures (e.g. NGT...)

Fowler's position cont'd...

□ **Contraindication**

- Comatose/unconscious patients
- Spinal injury
- Foot drop
- Head injury
- Shoulder dislocation

7. Lithotomy position

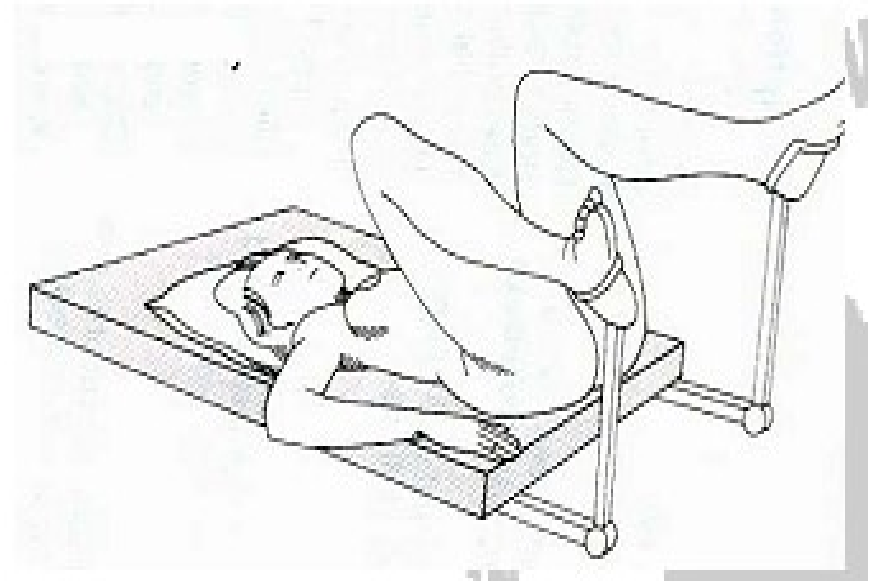
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□ Indication

- Pelvic medical examination and surgeries
- Delivery

□ Contraindication

- Spinal injury
- Breathing impairments



8. Trendelenburg position

➤ It is putting the patient in a flat on the back with the feet higher than the head by 15-30 degree



Trendelenburg position cont...

□ Indication

- ❖ Hypotension/shock
- ❖ Abdominal and gynecologic surgery
- ❖ Placing central venous line
- ❖ Surgical reduction of hernia

Trendelenburg position cont...

□ Contraindication

- ❖ Cervical –spine fracture
- ❖ Respiratory impairment/breathing difficulties
- ❖ Brain injury with increased intracranial pressure
- ❖ Spinal injury
- ❖ Cardiac patient (CHF)
- ❖ Breathing impairments
- ❖ Pressure sore (buttock, scrum, heel and shoulder)

Patient Transfer

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i. Transferring a Client from Bed to Stretcher

a. Three carrier lift

- ✓ It is alternative method of transferring helpless patients from bed to stretcher with three nurses or a nurse and two informed assistants.

□ Purposes

- To reduces client's anxiety and increases cooperation
- Decrease risk of client's falling
- Done for patients unable to move from bed to areas where procedures performed

□ Equipment

- Stretcher
- Pillow
- Clean glove

PROCEDURE

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1. Greet the patient and explain the procedure to the patient
2. Wash your hand.
3. Make sure that assistants are available
4. Adjust the head of bed to the flat position.
5. Put stretcher parallel to patient's head.
6. Each person must support one section of the patient's body-head, shoulders, and chest, hips and thighs and legs.
7. Slide your arms under the patient as far as possible and on signal, simultaneously roll the patient towards your chest.
8. Walking together, move patient from bed to stretcher
9. Observe the patient condition
10. Document the procedure.

b. Draw sheet method of transferring patient from bed to stretcher

- This is method of transferring patient from bed to stretcher using the sheet already under the patient as draw sheet.
- **Equipment**
 - Stretcher
 - Pillow
 - Lift sheet
 - Glove
 - Documentation format

Procedure

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1. Greet the patient and explain the procedure
2. Wash your hand and dry it
3. Done glove if necessary
4. Loosen bottom sheet beneath patient
5. Position stretcher next to and parallel to bed
6. Prepare stretcher and adjust to bed height
7. Adjust the draw sheet
8. Across stretcher and grasp sheet firmly at the patient's head, chest, hips and knees
9. Give direction
10. Slide the patient gently on to stretcher
11. Comfort the patient
12. Observe the patient condition
13. Document the procedure.

Transferring client from bed to stretcher with Maximum Assistance



Range of motion exercise/ROM

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- **Range of motion** reflects the extent to which a joint can move
- The ranges vary with each joint and are affected by several factors, including age, physical condition, and heredity.
- Range of motion exercise refers to activities aimed at improving movement of specific or group of joints

□ Purpose

- To maintain the current joint function
- To restore joint function that has been lost through disease or injury, or lack of use
- To maintain muscle tone and strength
- To prevent contractures
- To improve circulation

Types of ROM

ROM cont...

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- **Active:** movements of the joints independently by the client/patient on a nonfunctioning joints
- **Passive:** movement applied by a nurse or other person or passive motion machine on a pt's immobilized joint
- **Active-Resistive:** done by the client against a weight or force
- **Active-Assistive:** done by stronger arm and leg to weaker arm and leg
- **Progressive ROM:** use of weights (dumbbells, barbells, etc.)

□ Indication

- ❖ Unable to move joints
- ❖ Part of daily living activities

□ Contraindication

- ❖ Dislocation in specific joints
- ❖ Heart & respiratory disease and Swollen or inflamed joints or musculoskeletal injury

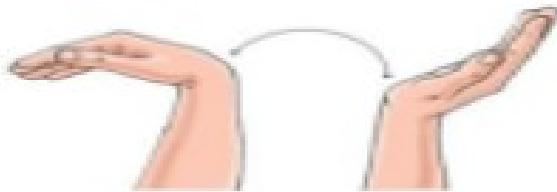
Active Range of Motion (AROM)

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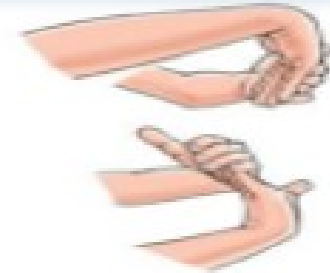
- Active range of motion is a flexibility technique commonly used by physical therapists as part of **rehabilitation programs**
- It is not actually a method of stretching but is used to promote normal motion.
- An e.g. is opening and closing your hand, making sure that you completely extend and flex all of the fingers each time.
- For the best results, a range of motion exercise should be performed 5 to 10 times, depending on comfort

AROM Cont'd..

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Wrist active range of motion: Flexion and extension



Wrist stretch



Forearm pronation and supination



Wrist flexion



Wrist extension



Grip strengthening



Forearm pronation and supination strengthening



Resisted elbow flexion and extension

PROCEDURE

1. Determine any restriction for active range of motion exercises.
2. Plan time for exercises to be done several times a day.
3. Identify the client.
4. Explain the procedure to the client.

5. Ankle.

- Write out the letters of the alphabet using sweeping capital letters.
- Do all letters three times.

6. Achilles tendon.

- In a seated position, wrap a towel around the forefoot and slowly stretch the Achilles tendon.

7. Knee.

- In a seated position with the knee slightly flexed, wrap a towel around the lower leg and slowly bring the ankle and foot toward the buttocks.

8. Hip.

- Lying down, bring one knee toward the chest.
- Repeat with the other knee.

9. Wrist or elbow.

- Using the unaffected hand, slowly stretch the affected hand or forearm in flexion and extension.

10. Shoulder flexion.

- With a wand or cane, use the unaffected arm to slowly raise the wand high above the head.

11.Shoulder lateral (external) rotation

- With a wand or cane, use the unaffected arm to slowly do **lateral (external) rotation** of the glenohumeral joint.

12.Shoulder medial (internal) rotation.

- With a wand or cane, use the unaffected arm to slowly do **medial (internal) rotation** of the glenohumeral joint.

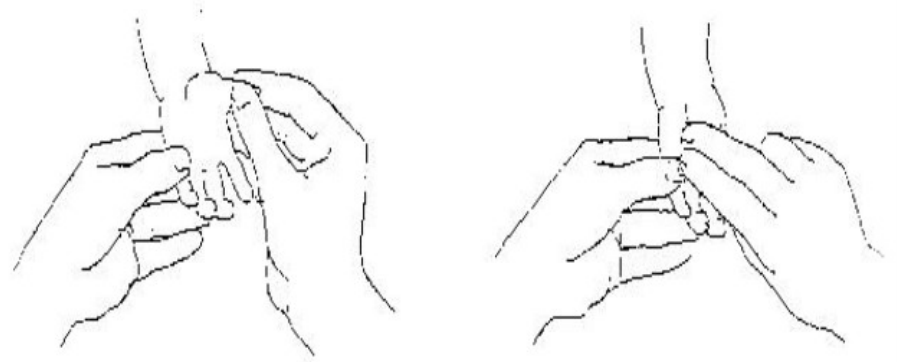
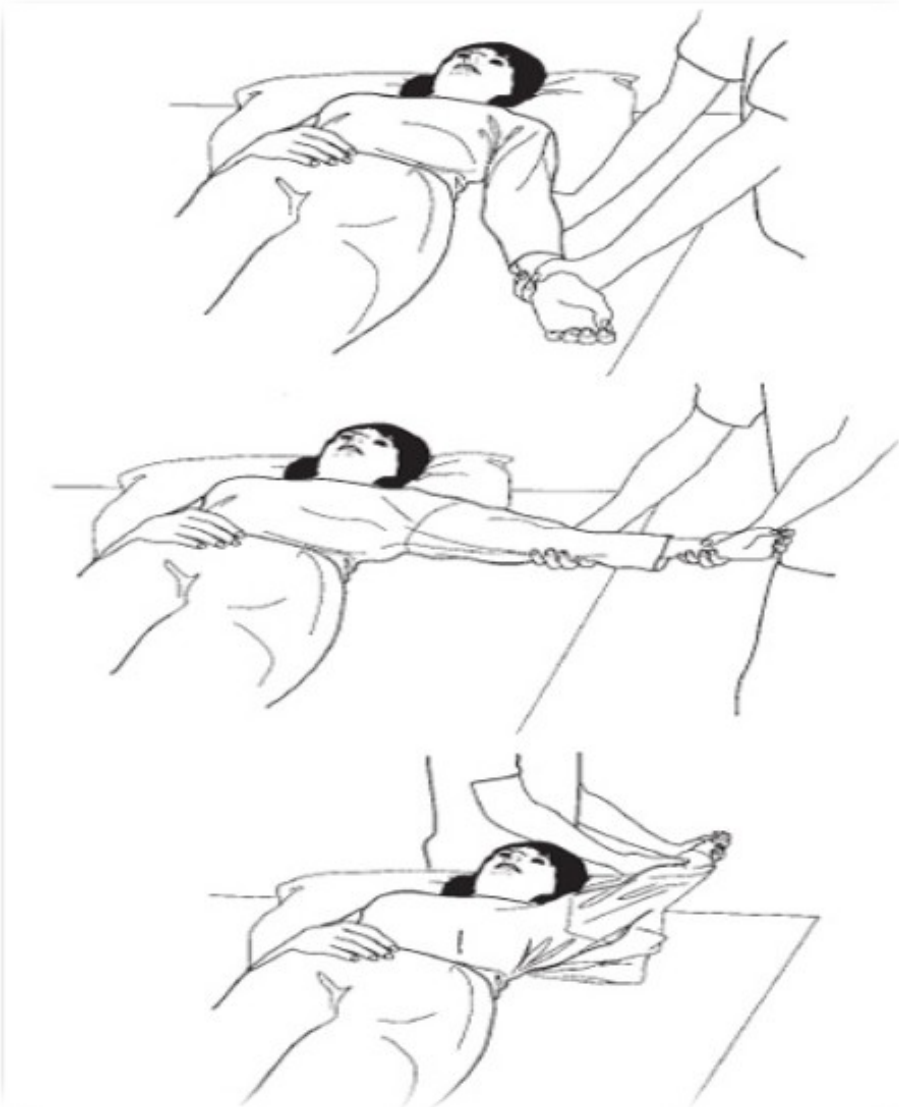
Passive range of motion

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(PROM)

- Passive range of motion exercises are done for a person by a helper.
- The helper does the range of motion exercises because the person cannot do them by himself.
- Without these exercises, blood flow and flexibility (moving and bending) of your joints can decrease.
- Joints, such as your knees and elbows, could become stiff and locked without range of motion exercises.

PROM Cont'd..



Precaution during PROM:

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1. Determine any restriction for passive range of motion exercises.
2. Plan time for exercises to be done several times a day.
3. Wash hands.
4. Identify the client.
5. Explain the procedure to the client.
6. Elevate the bed and place the client supine.
7. Start at the head and do exercise down each side of the body. Repeat each exercise 5 times.
8. For each exercise, support the appropriate joints.

Procedure of Passive ROM P

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1. Wash hands
2. Explain the procedure to the client
3. Adjust the bed to a comfortable height.
4. Select one side of the bed to begin PROM exercises.
5. Uncover only the limb to be exercised.
6. Support all joints during exercise activity.
7. Use slow, gentle movements when performing exercises.
8. Repeat each exercise three times.

PROM Cont'd...

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9. Stop if the client complains of pain or discomfort.

10. Begin exercise with the client's neck and work down ward.

11. Exercise the neck

- Flex, extend and rotate the client's neck.
- Support his or her head with your hands.

12. Exercise the client's shoulder and elbow.

- ▣ Support the client's elbow with one hand and grasp the client's wrist with your other hand.
- ▣ Raise the client's arm from the side to above the head.
- ▣ Perform internal rotation by moving the client's arm across his or her chest.
- ▣ Externally rotate the client's shoulder by moving the arm away from the client.
- ▣ Flex and extend the client's elbow.

PROM Cont'd...

100

13. Perform all exercises on the client's wrist and fingers

- Flex and extend the wrist.
- Abduct and adduct the wrist.
- Rotate the wrist.
- Flex and extend the client's fingers.
- Abduct and adduct the fingers.
- Rotate the thumb.

14. Exercise the client's hip and leg.

- Flex and extend the hip and knee while supporting the leg.
- Abduct and adduct the hip by moving the client's straightened leg toward you and then back to median position.
- Perform internal and external rotation of the hip joint by turning the leg inward and then outward.

PROM Cont'd...

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15. Perform exercises on ankle and foot

- Dorsiflex and plantar flex the foot
- Abduct and adduct the toes
- Evert and invert the foot

16. Move to the other side of the bed and repeat exercise.

17. Position and cover the client. Return the bed to low position.

18. Wash your hands.

19. Document completion of PROM exercise

Thank You So Much!!



Nursing is a Holistic Profession!!